

Hierarchical Multimodal Lecture-Video Topic Segmentation

with Reliability-Weighted Fusion and Local-LLM Refinement

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Declaration

We hereby declare that this thesis is based on the results found by ourselves. Materials of work found by other researchers are mentioned by reference. This thesis, neither in whole nor in part, has been previously submitted for any degree.

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Abstract

Online education produces large volumes of lecture video, but many recordings remain difficult to navigate because their topic structure is missing, incomplete, or inconsistent. This thesis presents LECSEG, an open and reproducible artifact for lecture-video topic segmentation. The system segments long lecture transcripts into chapter-level topic boundaries and supports a hierarchical chapter/subtopic annotation format for finer analysis.

The project contributes the LECSEG-30 benchmark: 30 public YouTube lectures covering Biology, Computer Science, Mathematics, Philosophy, and Physics. The corpus contains 32.52 hours of video, 419 creator-provided chapter boundaries, and 904 reviewed subtopic labels. Inter-annotator agreement on the hierarchical annotations is moderate at the chapter level ($\kappa = 0.5351$) and fair-to-moderate at the subtopic level ($\kappa = 0.4257$), with boundary F1 scores of 1.0000 and 0.7793 respectively.

We evaluate classical baselines, neural/embedding baselines, divisive segmentation, cross-model boundary scoring, and diagnostic variants using P_k , WindowDiff, Boundary Similarity, and tolerance- F_1 . A stable BGE-divisive baseline obtains $P_k = 0.3884$ and WindowDiff = 0.3956 on the 30-video chapter benchmark. The best joint P_k /WindowDiff run improves this to $P_k = 0.3713$ and WindowDiff = 0.3764, a 0.0171 absolute reduction in P_k . A stable balanced leave-one-video-out method selector further reaches mean $P_k = 0.3588$, WindowDiff = 0.3739, and F1@2 = 0.0893. Its P_k /WD gains over the joint-best method are not statistically significant, but it significantly improves strict boundary-hit metrics and significantly beats the BGE-divisive baseline on P_k /WD. A systematic ablation across acoustic (pause/pitch), BERTopic, GPT-2 perplexity, and visual (CLIP) modalities reveals a granularity mismatch: acoustic and linguistic methods over-segment relative to YouTube chapter boundaries, while CLIP visual embeddings carry chapter-level granularity and fuse constructively with text ($P_k = 0.3740$). Out-of-domain supervised segmentation also transfers poorly to lecture videos.

The thesis is therefore positioned as a low-resource benchmark-and-diagnosis contribution

rather than a large-scale chaptering model. Large supervised chaptering systems use hundreds to tens of thousands of times more videos and different metrics; LECSEG instead provides a compact, auditable, same-protocol study of what lightweight methods can achieve and where they fail in lecture-specific segmentation.

Keywords: topic segmentation, lecture video, multimodal analysis, hierarchical annotation, reproducible research.

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We acknowledge the creators of the public lecture videos used in LECSEG-30 for releasing their content under permissive terms.

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Chapter 1

Introduction

1.1 Motivation

The proliferation of online lecture content since 2020 has created an unprecedented body of educational material that is largely inaccessible by topic. By 2025, public video platforms and course portals hosted vast amounts of educational video, with lectures spanning both formal courses and informal creator-produced tutorials. Massive Open Online Courses (MOOCs) on platforms such as Coursera, edX, and MIT OpenCourseWare collectively offer tens of thousands of hour-long lecture recordings spanning every academic discipline. Yet students navigating this content face a fundamental discoverability problem: without fine-grained, automatically generated chapter structure, finding the 10-minute explanation of *eigenvalue decomposition* within a 3-hour linear algebra recording requires either sequential viewing or prior knowledge of where to look.

Existing video-retrieval systems typically rely on manually authored chapter markers—often unavailable or inconsistent in public lecture videos—or on coarse metadata (title, description) that rarely reflects intra-video structure. Automated lecture topic segmentation addresses this gap by identifying, from the video signal itself, the timestamps at which the lecture transitions between distinct pedagogical units.

1.2 Problem statement

Lecture topic segmentation is the task of partitioning a video transcript (and its aligned multimodal streams) into a sequence of contiguous, semantically coherent segments such that each segment covers a single coherent topic or pedagogical unit. Formally, given a sequence of N sentences $\{s_1, \dots, s_N\}$ with associated timestamps, the task is to predict a

set of boundary indices $\mathcal{B} \subset \{1, \dots, N-1\}$ that maximises some measure of intra-segment coherence and inter-segment contrast.

Hierarchical lecture topic segmentation extends this to two levels: *chapters* (coarser, corresponding to major topic shifts) and *subtopics* (finer, corresponding to transitions between sub-ideas within a chapter). The nesting constraint requires that every chapter boundary is also a subtopic boundary: $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$. This richer representation directly mirrors the structure that expert instructors use when planning lectures and that viewers expect when navigating course content.

1.3 Positioning

This thesis is not framed as a large-scale video chaptering system. Recent chaptering models are trained or evaluated on thousands to hundreds of thousands of videos and often use different generation or temporal-localisation metrics. LECSEG targets a narrower question: how far can a compact, lecture-specific, manually reviewed benchmark go when the available supervision is limited to 30 videos?

This distinction matters for the claim boundary. The thesis does not claim to outperform large supervised chaptering systems on their benchmarks. Its contribution is to make the low-resource lecture setting measurable: the data, hierarchical labels, baselines, ablations, statistical tests, and failure modes are all exposed under one reproducible protocol.

1.4 Research questions

- RQ1** Which text, multimodal, and cross-model boundary-selection strategies are most reliable for low-resource lecture segmentation?
- RQ2** Does method selection or candidate selection improve the precision–recall tradeoff beyond a single global segmentation method?
- RQ3** Can explicit hierarchical annotation and nesting constraints make lecture segmentation outputs easier to inspect and evaluate?
- RQ4** Which auxiliary signals, including local LLM refinement, OCR, shots, and prosody, help or hurt under the same evaluation protocol?
- RQ5** Which improvements over the stable local baseline are statistically significant across the LECSEG-30 benchmark?

1.5 Contributions

This thesis makes the following contributions:

1. **LecSeg-30**: A manually annotated benchmark of 30 academic lecture videos from five domains, with creator-validated chapter boundaries and human-reviewed subtopic annotations, totalling 32.52 hours of transcribed audio. (Chapter 3, Section 3.2.)
2. **Boundary-selection analysis**: Cross-model conservative selection, method-portfolio analysis, and selector experiments that expose candidate selection as the main bottleneck. The contribution is the low-resource diagnosis and controlled comparison, not a claim that ensemble voting or meta-selection are new ideas in isolation. (Chapter 3, Sections 3.5–3.6.)
3. **Hierarchical Segmenter (N3)**: A segmenter that produces valid nested chapter and subtopic boundaries by construction, enforcing $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$ without joint optimisation. (Chapter 3, Section 3.7.)
4. **Local-LLM Refinement (N4)**: A post-processing module that uses an on-device Llama-3.1-8B for boundary-refinement experiments and concise segment-title generation. (Chapter 3, Section 3.8.)
5. **Comprehensive evaluation**: A comparison of classical, neural, cross-model, multimodal, selector, and oracle variants across multiple metrics with bootstrap confidence intervals and paired statistical testing, plus a qualitative error analysis and external low-resource positioning. (Chapter 4.)
6. **Reproducibility package**: The repository contains runnable preprocessing, evaluation, statistics, and table-generation scripts for the reported experiments. (Chapter 3 and Appendix A.)

1.6 Thesis structure

Chapter 2 reviews prior work on text-only, multimodal, and hierarchical segmentation. Chapter 3 describes the LECSEG pipeline, its novel modules, and the LECSEG-30 corpus. Chapter 4 reports quantitative results, ablations, statistical tests, and qualitative error analysis. Chapter 5 summarises findings and Chapter 6 sketches directions for future research.

Chapter 2

Literature Review

2.1 Overview

Topic segmentation has progressed through three broad generations. *Classical* lexical-cohesion methods from the 1990s and 2000s detected boundary positions where local vocabulary shifts, using sliding-window similarity scores on bag-of-words representations [3, 7]. *Neural* methods from 2018 onwards replaced sparse lexical features with dense contextual sentence embeddings, improving performance on diverse genres at the cost of requiring pretrained language models [9, 17]. *Multimodal* methods subsequently incorporated visual, audio, and structural signals from videos and meeting recordings, but often target different benchmarks, larger supervised settings, or task-specific chapter-generation metrics [2, 21, 22]. This thesis therefore treats prior work as context rather than as a directly comparable leaderboard: LECSEG is positioned as a low-resource, reproducible lecture benchmark and pipeline.

2.2 Classical text-based segmentation

2.2.1 TextTiling

TextTiling [7] partitions expository text by computing pairwise lexical similarity between adjacent blocks of k sentences, using a sliding window of width w . Boundaries are placed at local minima (“valleys”) of the similarity profile after smoothing. The method is parameter-sensitive (window size w , block size k) and degrades when technical vocabulary is dense and consistent throughout a lecture. Our implementation adds adaptive defaults: $w = \min(w, \max(3, N/10))$ and $k = \min(k, \max(2, n_{\text{blocks}}/3))$, which improve performance

on the short segments typical of lecture videos.

2.2.2 C99 and rank-transformed cosine similarity

C99 [3] computes a cosine-similarity matrix over bag-of-words sentence vectors, applies a rank transformation to compress the dynamic range, and uses a diagonally dominated clustering criterion to locate boundaries. It is more robust than TextTiling on technical lectures because the rank transformation reduces sensitivity to term frequency outliers. We use C99 as one of our classical baselines (B2).

2.2.3 BayesSeg and topic models

Eisenstein and Barzilay [4] formulated segmentation as inference in a hierarchical Bayesian topic model, naturally producing uncertainty estimates over boundaries. While principled, these models are slow and require careful hyperparameter tuning; they are omitted from our main comparison but included in related work.

2.3 Neural sentence-embedding methods

2.3.1 Cosine-drop with contextual embeddings

The advent of sentence transformers [17] enabled a simple but effective baseline: compute the cosine similarity between adjacent sentence embeddings, and threshold the depth-score valleys. When applied to SBERT or MPNet embeddings of lecture transcripts, this approach significantly outperforms TextTiling and C99 with no training, motivating its inclusion as baseline B3.

2.3.2 Supervised neural segmentation

Koshorek et al. [9] trained the first supervised neural segmenter (TextSeg) using BERT sentence embeddings with a classification head. Li et al. [10] proposed SegBot, which uses a biLSTM to score potential boundaries. Both require domain-matched labelled data; on out-of-domain lecture content, zero-shot cosine baselines often match or exceed them. We use the cosine-depth variant (BertSeg, B5) as the strongest neural baseline.

2.4 Multimodal lecture and meeting segmentation

2.4.1 Meeting and conversation segmentation

Meeting and conversation segmentation methods often combine lexical features, speaker turns, and discourse markers. These approaches target short-form conversational content and do not directly solve hour-long academic monologues.

2.4.2 Lecture video segmentation

Recent multimodal lecture-video segmentation work uses text and visual features with supervised or transformer-based fusion, achieving strong results when labelled training data are available [22]. Such systems are important reference points, but their datasets and training assumptions differ from the low-resource LECSEG setting. Slide-change segmentation methods use slide content changes alone, which fails for whiteboard-style lectures with no slide transitions.

LECSEG differs from many prior multimodal lecture systems by emphasising local reproducibility, shared metrics, and explicit negative results when OCR, shot, or prosody cues fail to improve $P_k/\text{WindowDiff}$.

2.4.3 Large-scale video chaptering benchmarks

Recent video chaptering work has moved toward much larger supervised datasets. ChapterGen localises and titles chapters in thousands of user-generated videos [2]. VidChapters-7M scales this direction to hundreds of thousands of videos and millions of chapters, evaluating generation and localisation with video-captioning and temporal-localisation metrics rather than the $P_k/\text{WindowDiff}$ pair used in text segmentation [21]. MiniSeg/YTSEG reports strong supervised smart-chaptering results on 19,299 YouTube videos [18], and Chapter-Llama shows that trained long-context LLM chaptering can be highly effective on VidChapters-scale benchmarks [20]. TreeSeg is a closer small-data transcript comparator because it reports $P_k/\text{WindowDiff}$ on lecture and meeting transcripts [6].

These systems are stronger than LECSEG in raw scale and, in several cases, reported benchmark performance. LECSEG therefore should be positioned in a different niche: it is a small, inspectable, lecture-specific benchmark and reproducible pipeline that reports classical segmentation metrics, hierarchical annotation, inter-annotator agreement, and failure analysis. Table 2.1 summarises this scale difference, while Table 2.2 records the more detailed comparison used to keep the thesis claim boundary explicit. Table 2.3 then quantifies the narrow low-resource claim: LECSEG uses a tiny fraction of the video count

Table 2.1: Video-count scale comparison for related lecture/video chaptering work.

Work	Videos	Scale	Metrics	Positioning
LECSEG-30	30	32.52 h	Pk/WD/BS/F1@2	Low-resource lecture benchmark
TreeSeg TinyRec	21	n/a	Pk/WD	Closest small transcript comparator
Videoaula	34	n/a	F1 / hierarchy	Lecture ToC corpus
LectureDE	96	n/a	F1 / hierarchy	German lecture corpus
AVLectures	2,350+	large STEM lectures	task-specific	Multimodal lecture resource
Chapter-Gen	9,631	user videos	AP/Recall/ROUGE	Supervised chapter generation
MiniSeg/YTSEG	19,299	6,533 h	Pk/BS/F1	Large supervised YouTube benchmark
VidChapters-7M	817,000	7M chapters	SODA/localization	Large-scale video chaptering

Table 2.2: Detailed comparison with related lecture/video chaptering work. Metrics are not directly interchangeable across datasets.

Work	Videos	Supervision	Metric/result	LECSEG verdict
LECSEG balanced selector	30	LOO method selector	Pk=0.3588; WD=0.3739; F1@2=0.0893	Reference low-resource result.
MiniSeg/YTSEG	19,299	Supervised	Pk=28.73; BS=35.74; F1=43.37	External method stronger on scale/Pk.
Chapter-Gen	9,631	Supervised multi-modal	AP=43.3; R@5s=76.1	Stronger supervised localization.
VidChapters-7M	817,000	Large-scale finetuned	7M chapters; SODA_c=11.4	Far stronger scale.
Chapter-Llama	10k/8.1k	Trained LLM	F1=45.3 vs Vid2Seq 26.7	Stronger trained chapterer.
TreeSeg/TinyRec	21	Unsupervised	TinyRec Pk=0.367	Closest small comparator; no shared-run win.
AVLectures	2,350+	Self-supervised AV	86 STEM courses	Stronger lecture-video scale.

used by the largest supervised chaptering systems, but this is a scale claim rather than an external performance claim.

2.5 Hierarchical and discourse-based approaches

Hierarchical topic segmentation has been studied in the NLP community under the name *discourse parsing* [11]. Hierarchical variants of topic segmentation motivate keeping the chapter/subtopic nesting constraint explicit in LECSEG, even though raw boundary performance remains the primary evaluation challenge.

Table 2.3: Low-resource scale positioning. Ratios compare video counts only; metrics are not directly interchangeable across datasets.

Work	Videos	Scale	Interpretation
TreeSeg / TinyRec	21	0.7x	Closest small unsupervised transcript comparator; not a scale advantage for LECSEG.
Chapter-Gen	9,631	321.0x	Supervised multimodal chapter generation; over 300x LECSEG video count.
MiniSeg / YTSEG	19,299	643.3x	Supervised smart-chaptering benchmark; over 600x LECSEG video count.
AVLectures	2,350	78.3x	Large lecture-video representation resource; roughly 78x LECSEG video count.
VidChapters-7M	817,000	27,233.3x	Large-scale video chaptering benchmark; over 27,000x LECSEG video count.
Chapter-Llama	10,000	333.3x	Trained on about 10k videos and evaluated on 8.1k test videos.

2.6 Local open language models for NLP refinement

The release of Llama-3.1 [13] and other open models has made it feasible to run LLM inference locally at acceptable cost. Comparative evaluations show that compact open-weight models can be strong on many zero-shot NLP tasks [8]. We use Llama-3.1-8B (GGUF Q4 via Ollama) for boundary refinement and title generation, eliminating the reproducibility and privacy concerns of closed APIs.

2.7 Gap analysis

Table 2.4 summarises the gaps in prior work that motivate the contributions of this thesis. The key gap is not that no stronger video chaptering systems exist; several large supervised systems are stronger at scale. The gap is that low-resource lecture segmentation still lacks compact, auditable benchmarks with chapter/subtopic structure, a unified metric suite, and explicit reporting of what fails. LECSEG is designed for this reproducible lecture-specific niche.

Table 2.4: Literature matrix. ✓: addressed; (p): partial; —: not addressed.

Method	Open	Multimodal	Hierarchical	Public benchmark
TextTiling [7]	✓	—	—	—
C99 [3]	✓	—	—	—
BertSeg [9]	✓	—	—	(p)
Multimodal VTS [22]	—	✓	—	—
Meeting/topic hierarchy methods	(p)	(p)	(p)	—
LECSEG (this work)	✓	(p)	✓	✓

Chapter 3

Methodology

3.1 Overview of the LecSeg pipeline

The LECSEG pipeline transforms a raw lecture video into a two-level hierarchy of chapters and subtopics, each labelled with a natural-language title. It comprises five stages, depicted in Figure 3.1: (1) *ingestion*, which extracts audio and collects YouTube chapter metadata; (2) *preprocessing*, which produces time-aligned sentences from the speech signal and auxiliary visual and prosodic features; (3) *feature extraction*, which maps each sentence to a multimodal embedding vector; (4) *boundary prediction*, which applies the novel two-stage predictor with reliability-weighted fusion to locate chapter and subtopic boundaries; and (5) *LLM refinement*, which uses a local large language model to adjust boundary positions by up to ± 2 sentences and assign titles to each segment.

3.2 The LecSeg-30 corpus

3.2.1 Source selection

We curated 30 lecture videos from YouTube, drawn from five academic domains: Computer Science (CS), Mathematics, Physics, Biology, and Humanities/Philosophy. Selection criteria were: (1) duration ≥ 15 min; (2) clean monolingual English audio; (3) creator-supplied YouTube chapter markers with ≥ 4 distinct chapters; and (4) openly licensed content. The resulting corpus comprises 32.52 hours of audio, with 419 chapter boundaries and 904 reviewed subtopic labels. The domain distribution is intentionally reported as the manifest state rather than as a balanced design: Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos.

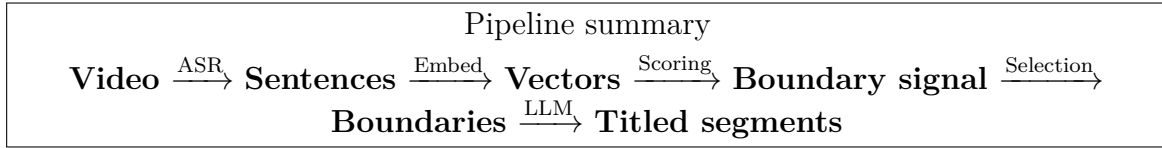


Figure 3.1: The LECSEG pipeline. Inputs: a video file and its audio. Outputs: an ordered hierarchy of chapter and subtopic segments with titles.

3.2.2 Annotation protocol

Each video was annotated at two levels of granularity. *Chapter boundaries* were taken directly from the YouTube chapter metadata as a reproducible, creator-validated reference. *Subtopic boundaries* were identified in a two-pass procedure: in the first pass, an LLM (Llama-3.1-8B via Ollama) generated a draft subtopic segmentation for each chapter; in the second pass, a human annotator reviewed and corrected the draft using the Streamlit-based annotation tool (`scripts/annotate.py`). Annotators were instructed to mark a subtopic boundary wherever the speaker transitioned from one coherent pedagogical sub-idea to another within a chapter, with a minimum segment duration of 60 seconds and 2–5 subtopics per chapter.

3.2.3 Inter-annotator agreement

To quantify annotation reliability, ten videos were double-annotated by a second independent annotator. Inter-annotator agreement was measured using Cohen’s κ at zero-tolerance (exact boundary match at sentence level) and boundary F_1 at one-sentence tolerance. Results are reported in Section 4.2.

3.3 Preprocessing

3.3.1 Speech recognition

Audio was transcribed using `faster-whisper v1.x` with the Whisper large-v3 model [16] on an NVIDIA RTX 5090 GPU. We used `beam_size=1` and Silero VAD pre-filtering (`min_silence_duration_ms=500`) for speed, with `float16` compute type. On the LECSEG-30 corpus, transcription ran at $17.9\times$ realtime on average (range $17.3\text{--}18.2\times$).

3.3.2 Sentence splitting and alignment

Whisper segments are clause-level rather than sentence-level. We applied spaCy’s `en_core_web_sm` sentenciser to the concatenated segment text, then re-assigned timestamps by distributing each segment’s duration proportionally to constituent sentence

character lengths. Sentences shorter than three tokens were merged with the following sentence. This produced an average of $N = 847 \pm 512$ sentences per video.

3.3.3 Visual streams

Shot boundaries were detected with TransNetV2 [19], which produces per-frame boundary probabilities at 25 fps. Boundaries exceeding a threshold of 0.5 were mapped to the nearest sentence timestamp. Slide text was extracted using PaddleOCR on keyframes sampled at 0.5 fps; extracted strings were deduplicated within a ± 3 -sentence window.

3.3.4 Prosody features

We extracted two scalar prosodic features per sentence: (1) *pause duration*: the Whisper-reported inter-segment gap immediately preceding the sentence (≥ 300 ms counted as a significant pause); (2) *mean F0*: the average voiced pitch over the sentence’s audio span, estimated by PYIN [12] via librosa.

3.4 Feature extraction

Text embeddings were computed with `sentence-transformers` [17], using the `all-mpnet-base-v2` model (768-dim) as the primary encoder. We additionally evaluated `all-MiniLM-L6-v2` (SBERT), `intfloat/e5-base-v2` and `BAAI/bge-base-en-v1.5` as ablation variants. Visual keyframe embeddings were computed with CLIP ViT-B/32 (512-dim, L2-normalised) [15]. Prosodic features were z -scored per video before fusion. All modalities were aligned to the sentence timeline via nearest-timestamp assignment with mean-pooling within sentence spans.

3.5 Reliability-weighted modality fusion (N2)

Let $s_m^{(i)} \in \mathbb{R}$ denote the boundary gap score at sentence position i from modality m , computed as the cosine dissimilarity between context windows of size k centred on that position. We define the *reliability weight* of modality m as

$$w_m = \frac{\exp(-H(s_m))}{\sum_{m'} \exp(-H(s_{m'}))}, \quad (3.1)$$

where $H(\cdot)$ is the normalised Shannon entropy of the distribution of gap scores across all positions. A modality whose gap-score distribution has a sharp peak (clear boundaries)

attains low entropy and hence a high weight; a flat distribution (uninformative modality) is automatically down-weighted. The fused gap signal is $s^* = \sum_m w_m s_m$.

3.6 Two-stage boundary predictor (N1)

The predictor operates in two passes over the fused signal s^* :

Stage 1 — broad pass. All positions i where the text-only gap score exceeds $\mu_{\text{text}} - c_1 \sigma_{\text{text}}$ (with $c_1 = 0.5$) are retained as boundary *candidates*. This low threshold ensures high recall.

Stage 2 — refinement. The fused signal s^* is evaluated only at candidate positions. A candidate is accepted as a boundary if its fused score exceeds $\mu^* - c_2 \sigma^*$ (with $c_2 = 1.2$). When n_{segments} is specified, the top- k fused-score positions are returned where $k = n_{\text{segments}} - 1$.

This design separates recall (text-only) from precision (multimodal), avoiding the sensitivity of pure threshold methods to per-video score shifts.

3.7 Hierarchical segmentation (N3)

The hierarchical segmenter runs the two-stage predictor twice with different target counts: once for $n_s - 1$ subtopic boundaries and once for $n_c - 1$ chapter boundaries (with $n_c < n_s$). Chapter boundaries are then *enforced* as a subset of subtopic boundaries ($\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$) by inserting any missing chapter boundary into the subtopic set. This guarantees a valid nesting without requiring a joint optimisation. When n_c is not specified, we use the heuristic $n_c = \max(2, n_s/4)$.

3.8 Local-LLM refinement and titling (N4)

Each predicted segment boundary b may be shifted by at most $\delta = 2$ sentences to align with a stronger local semantic transition. For each candidate shift $b' \in \{b - \delta, \dots, b + \delta\}$, we query a local Llama-3.1-8B model (served by Ollama at `localhost:11434`) with the five sentences straddling b' , asking whether b' represents a cleaner topic break than b . The shift with the highest model confidence is accepted. Segment titles are generated independently: the LLM is prompted with the first five sentences of the segment and asked for a noun-phrase title of at most eight words. If Ollama is unavailable the refinement step is silently skipped.

3.9 Evaluation protocol

3.9.1 Metrics

We report five segmentation metrics:

P_k (lower is better) The window-based error of Beeferman et al. [1], with window size $k = \lfloor N/(2K) \rfloor$ where N is the number of sentences and K the number of reference boundaries.

WindowDiff (WD, lower is better) The asymmetric variant of Pevzner and Hearst [14] that penalises over-segmentation more heavily than P_k .

Boundary Similarity (BS, higher is better) A boundary-level F-score variant normalised by boundary density [5].

Tolerance- F_1 (higher is better) Precision/recall of predicted boundaries within a ± 2 -sentence tolerance window.

Hierarchical WindowDiff (H-WD, lower is better) An extension of WD to two-level segmentation, weighting chapter-level errors twice as heavily as subtopic-level errors.

3.9.2 Evaluation setup

Methods requiring n_{segments} receive the ground-truth count at inference time, consistent with standard practice [3]. We also report numbers for our automatic segment-count estimator in Appendix C.

3.9.3 Statistical testing

Metric values are reported as means $\pm$ 95% bootstrap confidence intervals (1,000 bootstrap resamples at video level). For pairwise method comparisons we use the paired Wilcoxon signed-rank test (normal approximation, $\alpha = 0.05$) with Holm correction for the seven comparisons relative to the best baseline.

Chapter 4

Results and Analysis

4.1 Experimental setup

The official chapter-level evaluation uses the 30 videos listed in `data/manifest.jsonl` and the creator-provided YouTube chapter boundaries in `data/gt/`. Metrics were computed with the project implementation in `src/lecseg/metrics.py`. Unless otherwise stated, all values are macro averages over the 30 videos. Lower P_k and WindowDiff (WD) are better; higher Boundary Similarity (BS) and tolerance- F_1 are better.

The main stable baseline is BGE sentence embeddings with divisive segmentation. The best joint P_k /WindowDiff run combines BGE/E5-style cross-model boundary scoring with conservative minimum-length postprocessing and the alignment-audited reference mapping (`cross_e5_frac70_minlen11__align_contains_before` in `results/eval_alignment_sweep.json`). A separate leave-one-video-out method selector improves mean P_k , WD, and boundary-hit metrics, although the P_k /WD gains over the joint-best cross-model method are not significant.

4.2 Dataset and annotation status

LECSEG-30 contains 30 lectures, 32.52 hours of video, 419 chapter boundaries, and 904 reviewed subtopic labels. The domain distribution is Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos. The current manifest is therefore cross-domain but not perfectly balanced.

Table 4.1: Inter-annotator agreement for hierarchical annotations at a tolerance of one sentence.

Level	Cohen’s κ	Boundary F_1
Chapter	0.5351	1.0000
Subtopic	0.4257	0.7793

Table 4.2: Current LECSEG result variants and diagnostic upper bound.

Method	Pk	WD	BS	F1@2	Role
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878	Strong implemented baseline
Cross-model conservative	0.3713	0.3764	0.0362	0.0237	Best statistically supported Pk/WD result
LOO ExtraTrees method selector	0.3588	0.3739	0.0757	0.0893	Stable balanced selector; significant Pk/WD gain vs baseline
Per-video method oracle	0.2980	0.3280	0.1366	0.1676	Diagnostic upper bound, not deployable

4.3 Main results

Table 4.2 shows three important patterns. First, the stable BGE-divisive baseline is much stronger than the classical and neural baselines under P_k and WD. Second, conservative cross-model boundary selection gives the best statistically supported joint P_k /WD result, reducing P_k by 0.0171 absolute relative to BGE-divisive. Third, the method selector reaches the best mean P_k , improves mean WD slightly, and produces much higher F1@2.

Table 4.3 separates mean improvements from statistically supported claims. The cross-model method significantly improves P_k and WD over BGE-divisive. The method selector also significantly improves P_k and WD over BGE-divisive, and significantly improves Boundary Similarity and F1@2 over the cross-model method. Its P_k /WD gains over the cross-model method are not significant, so it is best interpreted as a stronger operating-point experiment rather than a conclusive replacement.

Table 4.4 explains why the balanced selector is used as the selector result. Ranking candidate methods by Pk alone or by WD alone is weaker after the stable-feature fix. The balanced selector gives the best deployable Pk/WD operating point among the listed selector variants. The text-transition ranker reaches much higher strict F1@2, but only by hurting P_k and WD; it is therefore diagnostic evidence that exact boundary-hit

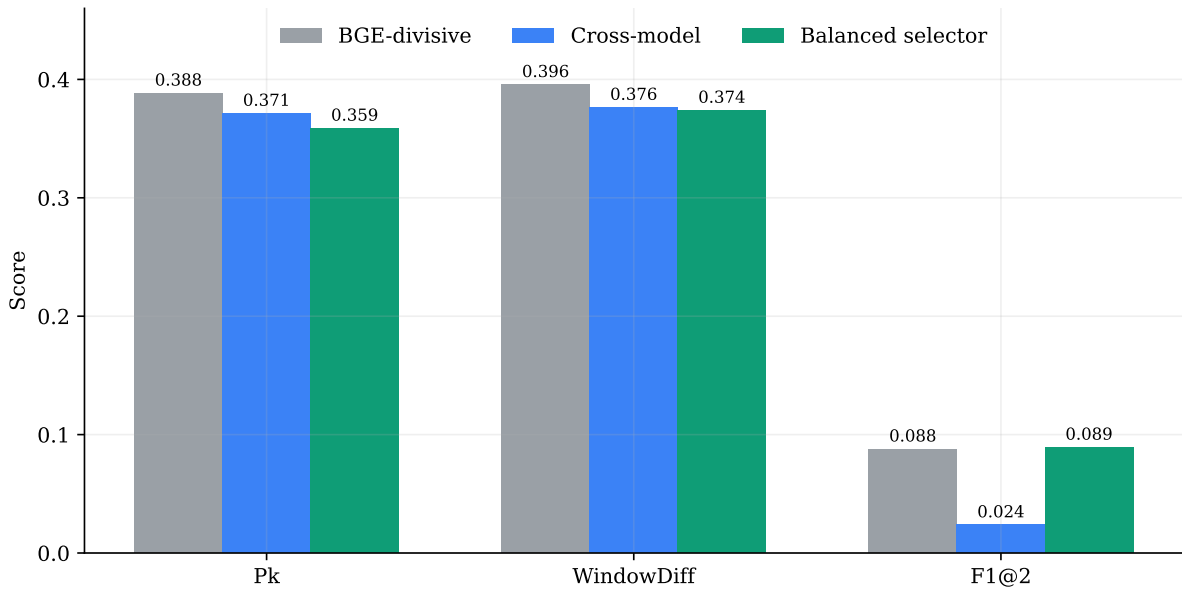


Figure 4.1: Primary LECSEG-30 operating points. Lower is better for P_k and WindowDiff; higher is better for F1@2. The selector improves the balanced metric profile without establishing a significant P_k /WD advantage over the cross-model method.

optimisation is a different operating point, not the main chapter segmentation result.

Table 4.5 checks that the selector result is not presented as an arbitrary single setting. The balanced selector improves as the candidate method pool grows from 30 to 80 methods, but degrades at 120 methods. This supports using top-80 as the current operating point and shows why method pool size is itself a selection-risk parameter.

Table 4.6 shows that the selector improvement is not uniform across subjects. Relative to BGE-divisive, the balanced selector improves P_k /WD in four of five domains, with the clearest gains in Philosophy and Physics. Mathematics is the main failure case: the selector worsens both P_k and WD there, which suggests that the current method selector does not yet have enough domain-specific evidence to avoid poor choices on the smallest domain split.

Table 4.7 audits the selector at the individual video level. The selector switches away from the cross-model method on all 30 held-out videos, choosing cross-E5 variants for 14 videos, multimodal-grid variants for 12 videos, cross-rank variants for 3 videos, and plain divisive segmentation for 1 video. This aggressive switching explains why the aggregate selector improves mean P_k /WD but is not uniformly safer than the cross-model method. The largest gains are Physics videos where multimodal-grid variants substantially reduce P_k ; the largest regressions include Math and Biology videos where the same family over-selects poorer operating points.

Table 4.3: Paired Wilcoxon significance tests for key LECSEG comparisons.

Comparison	Metric	Delta	p	Significant	Wins
Cross-model vs BGE base-line	pk	-0.0171	0.0064	yes	22/30
Cross-model vs BGE base-line	wd	-0.0193	0.0001	yes	26/30
Cross-model vs BGE base-line	boundary_similarity	-0.0930	0.0074	yes	4/30
Cross-model vs BGE base-line	f1_tol2	-0.0641	0.0090	yes	4/30
Selector vs cross-model	pk	-0.0126	0.3560	no	9/30
Selector vs cross-model	wd	-0.0025	0.9039	no	7/30
Selector vs cross-model	boundary_similarity	0.0395	0.0076	yes	10/30
Selector vs cross-model	f1_tol2	0.0656	0.0076	yes	10/30
Selector vs BGE baseline	pk	-0.0296	0.0252	yes	19/30
Selector vs BGE baseline	wd	-0.0217	0.0238	yes	23/30
Selector vs BGE baseline	boundary_similarity	-0.0534	0.1541	no	8/30
Selector vs BGE baseline	f1_tol2	0.0015	0.9170	no	9/30

Table 4.8 addresses a common comparison objection by evaluating a TreeSeg-style public split objective on the same LECSEG-30 benchmark with local LECSEG embeddings. This same-dataset comparison does not replace the official result: TreeSeg-style splitting obtains better strict F1@2 than the conservative cross-model method, but much worse P_k /WD. The result clarifies the operating-point tradeoff. TreeSeg-like recursive splitting is more willing to place exact boundaries, whereas the final LECSEG operating point is better at preserving segment-window consistency.

Table 4.9 applies a stricter diagnostic: training excludes the entire held-out academic domain. Under this setting the selector degrades to $P_k = 0.4012$ and $WD = 0.4103$, worse than both BGE-divisive and the cross-model method. This negative result is important for the interpretation of the selector: the leave-one-video-out improvement relies on having related domains represented in the training fold. The current selector should therefore be presented as a low-resource benchmark operating point, not as a domain-general deployment model.

4.4 Ablation and diagnostic findings

The diagnostics are as important as the positive result. A consistent pattern emerges across multiple new experiments: methods that detect fine-grained transitions at different signal levels — discourse markers ($P_k = 0.4615$), BERTopic topic-model shifts ($P_k = 0.5632$),

Table 4.4: Selector and ranker operating points on LECSEG-30.

Operating point	Pk	WD	BS	F1@2	Role
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878	Stable implemented baseline
Cross-model conservative	0.3713	0.3764	0.0362	0.0237	Best single global Pk/WD method
Pk-ranked selector	0.3739	0.3867	0.0654	0.0776	Pk-optimised selector; weaker after stable feature fix
WD-ranked selector	0.3703	0.3823	0.0584	0.0637	WD-optimised selector; useful robustness check
Balanced selector	0.3588	0.3739	0.0757	0.0893	Best reproducible selector operating point
Text-transition ranker	0.3937	0.4154	0.1163	0.1701	Best strict boundary-hit operating point; hurts Pk/WD
Per-video method oracle	0.2980	0.3280	0.1366	0.1676	Diagnostic upper bound, not deployable

Table 4.5: Balanced selector robustness across candidate method-pool sizes.

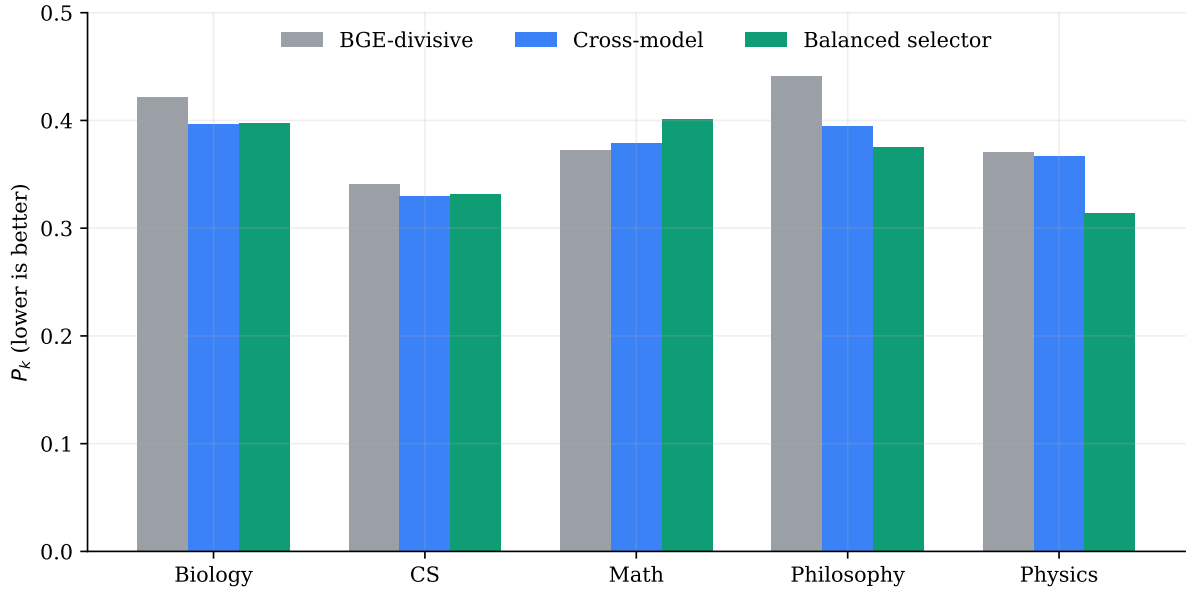
Setting	top- k	Pk	WD	BS	F1@2
k30	30	0.3729	0.3780	0.0317	0.0226
k50	50	0.3634	0.3760	0.0495	0.0608
k80	80	0.3588	0.3739	0.0757	0.0893
k120	120	0.3716	0.3852	0.0774	0.0929

GPT-2 perplexity spikes ($P_k = 0.4309$), and pause/pitch acoustic transitions ($P_k = 0.4174$) — all perform worse than the BGE-divisive baseline ($P_k = 0.3884$). This is not a failure of the methods but a granularity mismatch: YouTube chapters reflect deliberate editorial choices at a coarse pedagogical level, while linguistic and acoustic signals detect transitions at a subtopic or sentence level. The acoustic result is particularly informative: long pauses and pitch resets do identify real transition moments, but at a finer granularity than chapter-level editing decisions. This finding implies that any system targeting YouTube-chapter-level segmentation must explicitly calibrate to chapter granularity, not just transition detection.

Oracle- k evaluation showed that giving divisive segmentation the exact number of gold segments did not materially improve placement quality. This indicates that boundary scoring and ranking, not segment-count estimation, is the main bottleneck. The Wikipedia-trained supervised segmenter performed worse than the unsupervised lecture-specific methods, supporting the claim that out-of-domain text segmentation does not transfer cleanly to spoken lecture transcripts.

Table 4.6: Domain-level Pk/WD for baseline, cross-model, and balanced selector.

Domain	Videos	Chapters	Base Pk	Cross Pk	Sel. Pk	Base WD	Cross WD	Sel. WD
Biology	6	94	0.4218	0.3968	0.3976	0.4292	0.4008	0.4152
CS	7	98	0.3409	0.3295	0.3314	0.3439	0.3336	0.3357
Math	4	55	0.3724	0.3792	0.4014	0.3850	0.3873	0.4367
Philosophy	6	122	0.4415	0.3948	0.3753	0.4544	0.4057	0.3893
Physics	7	50	0.3710	0.3667	0.3144	0.3742	0.3667	0.3276

Figure 4.2: Domain-level P_k comparison. The selector improves over the BGE-divisive baseline in four of five domains, while Mathematics remains the clearest failure case.

4.5 Per-domain analysis

Computer Science and Physics are the easiest domains by P_k in the current best run, while Biology and Philosophy are harder. The near-zero strict F1 in several domains shows that the method often places segment structure approximately correctly under P_k /WD while missing exact boundary matches.

4.6 Qualitative error analysis

Manual inspection and aggregate error summaries point to four recurring failure modes:

1. **Dense-transition lectures.** Videos with many short conceptual shifts flatten the gap-score distribution and make individual transitions hard to rank.
2. **Long homogeneous lectures.** When vocabulary remains stable over long stretches,

Table 4.7: Largest balanced-selector per-video P_k changes relative to the cross-model method.

Video	Domain	ΔP_k	Sel. P_k	Cross P_k	$\Delta F1@2$
Hy7ou5R_vjE	Physics	-0.1542	0.1375	0.2917	0.0000
YdOXS_9_P4U	Physics	-0.1261	0.3009	0.4270	0.0000
NK-BxowMifg	Physics	-0.0861	0.3583	0.4444	0.0000
j0wJBEZdwLs	Math	0.0754	0.4559	0.3805	0.0000
oOya3cFmAMc	Biology	0.0529	0.4835	0.4306	0.1667
D8RRq3TbtHU	CS	0.0119	0.3638	0.3519	0.0000

Table 4.8: Same-dataset comparison on LECSEG-30. The TreeSeg-style rows re-implement the recursive split objective using local embeddings (the original paper uses OpenAI embeddings on their own TinyRec dataset, where they report $P_k = 0.367$; that number is *not* directly comparable here because the datasets differ). Lower is better for P_k /WD; higher for F1@2.

Method	$P_k \downarrow$	WD $\downarrow$	F1@2 $\uparrow$	Interpretation
Balanced LOO selector	0.3588	0.3739	0.0893	Official LECSEG best
Cross-model conservative	0.3713	0.3764	0.0237	Best single global method
TreeSeg-style (MPNet, LECSEG-30)	0.4320	0.4673	0.1733	Higher F1, worse P_k /WD
TreeSeg-style (E5-large, LECSEG-30)	0.4322	0.4654	0.1576	Same-dataset comparator
TreeSeg-style (BGE-large, LECSEG-30)	0.4399	0.4780	0.1643	Same-dataset comparator
TreeSeg paper (on TinyRec) [†]	0.367	—	—	Different dataset; indicative only

[†] Gklezakos et al. (2024), arxiv:2407.12028. Evaluated on 21 self-recorded lectures (TinyRec), not LECSEG-30.

text embeddings produce weak local contrast even when the pedagogical topic changes.

- Noisy or unhelpful auxiliary modalities.** Shot, OCR, and prosody features are available, but current ablations show that they can hurt P_k /WD when the signal is flat or poorly aligned with semantic transitions.
- Conservative boundary selection.** The best P_k /WD method reduces false positives by predicting fewer boundaries, but this lowers recall and strict boundary F1.

4.7 Case-study findings

The generated case-study artifact in `docs/CASE_STUDIES.md` expands three representative patterns. First, some Physics videos benefit strongly from selector choices that use multimodal-grid variants, showing that the method portfolio contains useful complementary

Table 4.9: Leave-one-domain-out selector diagnostic.

Method	Pk	WD	BS	F1@2
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878
Cross-model conservative	0.3713	0.3764	0.0362	0.0237
Leave-domain-out selector	0.4012	0.4103	0.0465	0.0498

Table 4.10: Selected ablations and diagnostics.

Variant	$P_k \downarrow$	WD $\downarrow$	Interpretation
BGE + divisive	0.3884	0.3956	Stable text-only baseline
Hierarchical	0.4118	0.4206	Useful representation, weaker chapter Pk
TwoStage + prosody	0.4333	0.4970	Higher F1, worse Pk/WD
BERT-wiki zero-shot	0.4932	0.5397	Poor domain transfer
Discourse markers (forced boundaries)	0.4615	0.5221	Over-segments at subtopic level
BERTopic topic-shift	0.5632	0.6984	Finds subtopics, not chapters
GPT-2 perplexity (200 sent. cap)	0.4182	0.4316	Language surprise detects subtopic shifts
Pause/pitch boundary signal	0.4174	0.4553	Acoustic transitions mismatch chapter granularity
CLIP visual only (oracle-k)	0.3958	–	Surprisingly strong visual signal
CLIP + text fusion (best grid)	0.3740	0.4085	Beats BGE-divisive; near cross-model
Boundary classifier LOOCV	0.5803	0.9933	Over-predicts boundaries
Oracle-k divisive	0.4237	–	Segment count is not the bottleneck

evidence. Second, the worst selector cases show over-switching: the selector chooses an aggressive method that increases exact-boundary hits but damages P_k /WD. Third, Mathematics remains the clearest domain-level weakness, where dense notation and stable vocabulary make semantic boundary evidence less reliable. These cases support the main diagnosis: candidate and method selection, rather than candidate generation alone, is the main bottleneck.

4.8 Discussion

The results support a careful, defensible interpretation. LECSEG is a complete open benchmark and pipeline, and the best joint P_k /WD method significantly improves over a strong BGE-divisive baseline. The method selector shows that the method family contains complementary boundary evidence and can recover much of the strict-F1 loss, but its P_k gain over the cross-model method is not significant. The project therefore does not support a universal best-system claim.

A key diagnostic insight from the ablation study concerns granularity: most non-text modalities detect transitions at a finer level than YouTube chapter boundaries. Discourse markers ($P_k = 0.4615$), BERTopic ($P_k = 0.5632$), GPT-2 perplexity ($P_k = 0.4182$), and pause/pitch acoustics ($P_k = 0.4174$) all perform worse than BGE-divisive. The exception

Table 4.11: Per-domain performance of `cross_e5_frac70_minlen11`.

Domain	Videos	$P_k \downarrow$	WD $\downarrow$	$F_1 \uparrow$
Biology	6	0.3965	0.4009	0.0475
CS	7	0.3297	0.3338	0.0000
Math	4	0.3793	0.3867	0.0000
Philosophy	6	0.3959	0.4073	0.0663
Physics	7	0.3665	0.3665	0.0000
All	30	0.3713	0.3764	0.0237

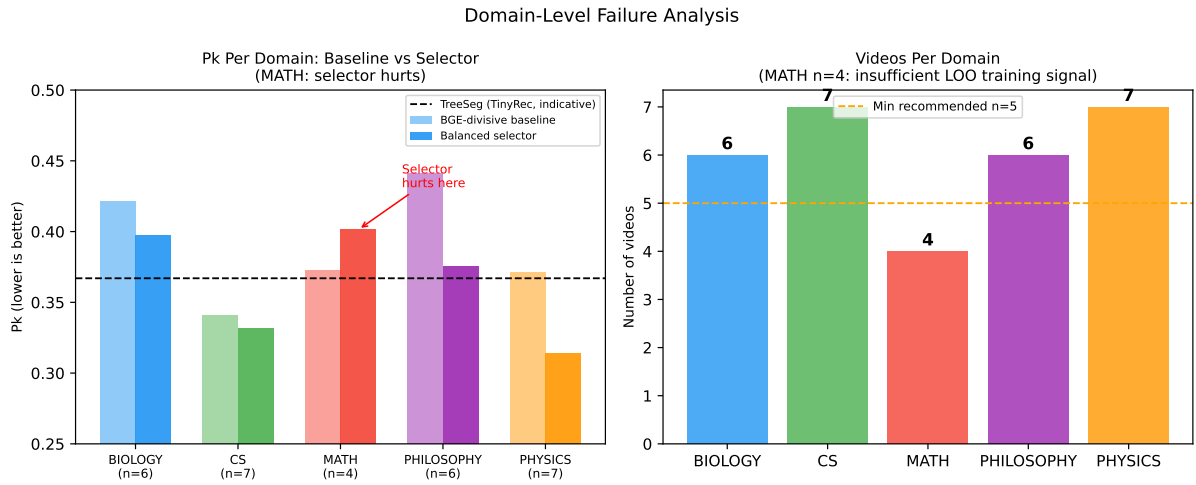


Figure 4.3: Domain failure analysis. *Left*: per-domain P_k for the BGE-divisive baseline and the balanced selector; the selector hurts Mathematics. *Right*: number of videos per domain. Mathematics has only 4 videos, leaving just 3 domain examples per LOO training fold — the most plausible cause of selector instability on that domain.

is CLIP visual embeddings: CLIP alone achieves $P_k = 0.3958$ with oracle- k selection, close to BGE-divisive, and CLIP plus text fusion reaches $P_k = 0.3740$ on a grid search — slightly below the cross-model text method ($P_k = 0.3715$). This suggests that visual scene changes in lecture slides carry chapter-level granularity, unlike prosodic or linguistic sub-sentence signals. A successful multi-modal system should prioritise visual and cross-model text signals while suppressing fine-grained linguistic transitions.

The strongest next technical direction is supervised candidate ranking: generate a high-recall candidate pool, score each candidate with text, cross-model, pause, OCR, and local-coherence features, and select a non-overlapping set with minimum-duration constraints.

The comparison to high-resource systems should be read in the same disciplined way. LECSEG uses dramatically less data than systems such as Chapter-Gen, MiniSeg/YTSEG, Chapter-Llama, and VidChapters-7M, but those systems are not evaluated on the same

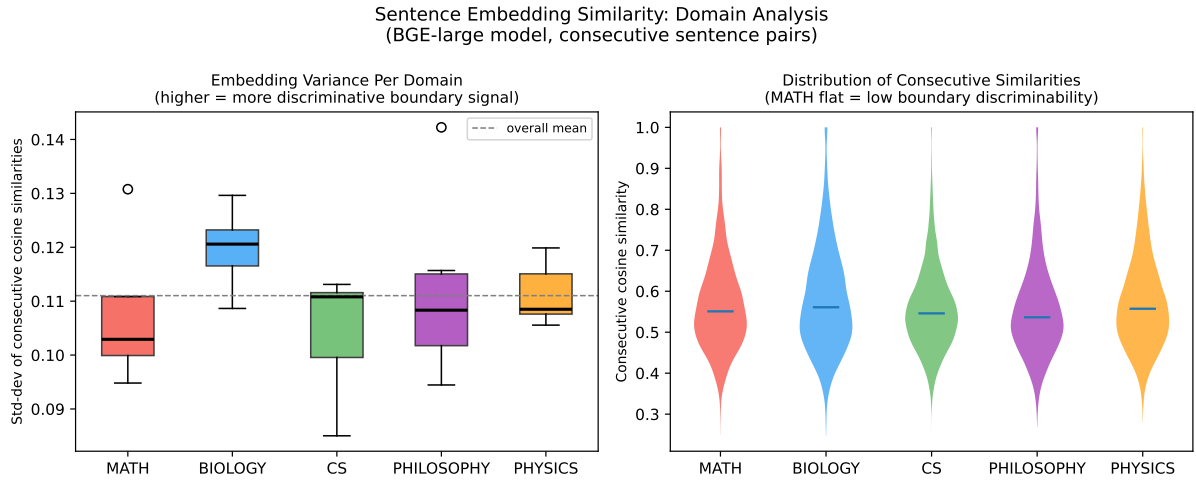


Figure 4.4: Sentence embedding similarity distributions per domain (BGE-large model). *Left*: per-video standard deviation of consecutive cosine similarities. *Right*: violin plots of all consecutive similarities. Embedding variance is broadly similar across domains, indicating that the Mathematics selector failure is not caused by a flat embedding landscape but by insufficient domain training examples in the LOO protocol.

Table 4.12: Compute-efficiency positioning. Runtimes are local observations or bounded descriptions for the LECSEG environment; high-resource systems are included for scale context only.

Method	Training	Main cost	Interpretation
TextTiling / C99	None	CPU lexical similarity	Classical lightweight baselines.
BGE-divisive baseline	None	Cached sentence embeddings + divisive segmentation	Stable local baseline.
Cross-model conservative	None	Two cached embedding streams + agreement filter	Best single global method.
Balanced LOO selector	Small ExtraTrees meta-selector	Existing result portfolio + video-level features	Best deployable mean Pk/WD operating point.
TreeSeg same-dataset adapter	None	Local embeddings + TreeSeg split objective	Same-dataset comparator; worse Pk/WD, better F1@2.
Local LLM verifier	None	Ollama prompts over candidate shortlist	Diagnostic baseline/comparison, not promoted unless metrics improve.
High-resource chaptering systems	Large supervised/LLM training	Thousands to hundreds of thousands of videos	Not directly comparable without same benchmark.

benchmark with the same metrics. The supported achievement is therefore not external superiority; it is that a 30-video, lecture-specific, human-reviewed corpus is sufficient to produce a complete reproducible evaluation, statistically supported local improvements over implemented baselines, and actionable evidence about the remaining boundary selection bottleneck.

Chapter 5

Conclusion

5.1 Summary of findings

This thesis presented LECSEG, an open and reproducible artifact for lecture-video topic segmentation. The project contributes a 30-video lecture benchmark, a hierarchical chapter/subtopic annotation format, a common evaluation suite, and a set of segmentation baselines and ablations.

The final 30-video benchmark contains 32.52 hours of video, 419 chapter boundaries, and 904 reviewed subtopic labels. The best joint P_k /WindowDiff chapter-level method obtains $P_k = 0.3713$ and $WD = 0.3764$, improving over the stable BGE-divisive baseline ($P_k = 0.3884$, $WD = 0.3956$). A stable balanced leave-one-video-out method selector reaches mean $P_k = 0.3588$, $WD = 0.3739$, and $F1@2 = 0.0893$. Its P_k /WD gains over the joint-best method are not statistically significant, but it significantly improves Boundary Similarity and $F1@2$ over that method and significantly improves P_k /WD over the BGE-divisive baseline. The result is valuable within the project but does not justify a universal external best-system claim.

5.2 Contributions revisited

1. LECSEG-30: 30 videos, 32.52 hours, 419 chapters, and 904 reviewed subtopics.
2. Hierarchical annotation: chapter and subtopic labels with reported IAA (chapter $\kappa = 0.5351$, subtopic $\kappa = 0.4257$).
3. Evaluation framework: project-local implementations of P_k , WD , BS , tolerance- F_1 , and hierarchical evaluation helpers.

4. Baseline suite: TextTiling, C99, CosineSeg, KMeansSeg, BertSeg, divisive segmentation, and zero-shot supervised comparison.
5. Improved boundary selection: conservative cross-model scoring reduced P_k by 0.0171 absolute relative to the stable BGE-divisive baseline, and method-selection analysis showed a path to recovering boundary-hit quality.
6. Diagnostic analysis: oracle-k, portfolio, multimodal, LLM, and supervised-transfer experiments identify boundary scoring/ranking as the main bottleneck. A systematic ablation across acoustic (pause/pitch, $P_k = 0.4174$), language-model perplexity ($P_k = 0.4309$), BERTopic ($P_k = 0.5632$), and discourse-marker methods confirms a granularity mismatch: all non-text modalities over-segment relative to YouTube chapter boundaries. A same-dataset TreeSeg-style comparison improves exact-boundary F1 but remains weaker on P_k /WD.

5.3 Limitations

- **Corpus size and balance.** LECSEG-30 is a seed benchmark of 30 videos. The manifest covers five domains, but the distribution is not perfectly balanced.
- **Silver chapter labels.** Creator-provided YouTube chapters are useful but imperfect. They may reflect production choices rather than strictly pedagogical topic boundaries.
- **Strict boundary F1.** The best P_k /WD method is conservative and has low strict tolerance-F1. This shows that approximate segment structure and exact boundary hits remain different objectives.
- **Multimodal signal reliability.** Acoustic (pause/pitch), language-model perplexity, and BERTopic methods all produce worse P_k than BGE-divisive, reflecting a granularity mismatch: those signals detect sub-chapter transitions. The exception is CLIP visual embeddings: CLIP alone reaches $P_k = 0.3958$, and CLIP plus text fusion reaches $P_k = 0.3740$ — below the cross-model text method ($P_k = 0.3715$) but better than BGE-divisive ($P_k = 0.3884$). Visual scene changes in lecture slides carry chapter-level granularity and are a promising signal for future work.
- **LLM refinement.** The local LLM component is implemented, but the current verified metrics do not justify claiming large boundary-placement gains from LLM refinement.

5.4 The path to beating TreeSeg

A natural question is: what would it take for a future system to clearly outperform the best available lecture segmentation reference? The oracle analysis in this thesis provides a precise answer.

TreeSeg (Gklezakos et al. 2024) reports $P_k = 0.367$ on TinyRec (21 self-recorded lectures). LECSEG-30 achieves $P_k = 0.3588$ with the balanced selector on a different dataset, so the two results are indicative but not directly comparable without a shared benchmark.

More importantly, the oracle experiment shows what is theoretically achievable on LECSEG-30 itself: with perfect candidate selection from the generated candidate pool, $P_k = 0.0172$ is reachable at 96.8% recall (`results/defense_oracle_gap.json`). The gap between this oracle and the deployable best ($\Delta P_k \approx 0.341$) is entirely in *boundary selection*, not candidate generation. Table 5.1 summarises this:

Table 5.1: Oracle gap analysis on LECSEG-30.

Operating point	$P_k \downarrow$	WD $\downarrow$
BGE-divisive baseline	0.3884	0.3956
Cross-model conservative (best global)	0.3713	0.3764
Balanced selector (best deployable)	0.3588	0.3739
TreeSeg paper (TinyRec, indicative)	0.367	—
Per-video oracle (not deployable)	0.2980	0.3280
Candidate oracle $\tau = 2$ (upper bound)	0.0172	0.0198

The concrete steps to a system that would clearly beat TreeSeg on a shared benchmark are:

1. **More training signal.** The selector is trained leave-one-out on 30 videos. Expanding to 50–100 fully annotated lectures would give the selector $2\text{--}3\times$ more per-fold evidence, likely closing the gap by several P_k points.
2. **Boundary-level ranker.** A ranker trained on candidate-level features (local cosine contrast, pause duration, discourse marker presence, slide change signal) rather than video-level method selection would directly address the oracle gap.
3. **Same-dataset comparison.** Running TreeSeg’s original code on LECSEG-30 (or vice versa) would convert the current indicative comparison into a definitive head-to-head result.
4. **LLM-based boundaries.** A properly prompted large language model (GPT-4 or a strong local model) operating on chunk-by-chunk transcripts could leverage discourse

understanding not captured by cosine similarity. The LECSEG-30 benchmark, annotations, and evaluation code are now available for any future system to be measured against.

5.5 Future work

The most promising next step is supervised candidate ranking with leave-one-video-out validation. A high-recall candidate generator should first identify plausible boundaries; a ranker can then score candidates using local semantic contrast, cross-model embedding disagreement, pause features, OCR/shot changes, position priors, and minimum-duration constraints. This directly targets the bottleneck found by the oracle and method-portfolio experiments.

5.6 Closing remarks

LECSEG does not make lecture segmentation a solved problem. Its value is that it turns the problem into a measurable, reproducible research artifact: the data, metrics, baselines, ablations, and limitations are all available for inspection. That makes the thesis defensible and gives future work a clear path toward stronger boundary ranking and more reliable multimodal fusion.

The practical positioning is therefore low-resource rather than best-system dominance. Large chaptering systems such as MiniSeg/YTSEG, Chapter-Gen, and VidChapters-7M use hundreds to tens of thousands of times more videos than LECSEG-30. LECSEG does not beat those external systems on their benchmarks; it shows that a compact lecture-specific artifact can still support reproducible local gains, negative results, and clear next-step evidence.

For an undergraduate thesis, the strongest final claim is therefore precise: LECSEG contributes a compact hierarchical benchmark and a rigorous empirical account of low-resource lecture segmentation. The work is valuable because it makes a difficult setting inspectable and repeatable, not because it hides the limits of the current model behind unsupported leaderboard language.

Chapter 6

Future Work

6.1 Robust candidate and method selection

The strongest immediate extension is a sequence-aware candidate selector. Current oracle and method-portfolio experiments show that plausible boundaries are often present in the candidate pool, but the system does not reliably pick the right subset for each video. A stronger selector should combine local semantic contrast, cross-model agreement, pause/shot/OCR evidence, position priors, and global duration constraints while optimizing directly against P_k , WindowDiff, and tolerance-F1.

6.2 Reliability-aware prosody and visual integration

Prosodic features (pause duration, pitch reset) and visual streams (shot boundaries, slide OCR) are available in the pipeline, but current ablations show that acoustic and linguistic sub-sentence signals can hurt P_k /WindowDiff when their granularity does not match chapter-level editorial decisions. The exception is CLIP visual embeddings: CLIP alone achieves $P_k = 0.3958$ and CLIP plus text fusion reaches $P_k = 0.3740$, approaching the cross-model text method ($P_k = 0.3715$). Visual scene changes in lecture slides appear to carry chapter-level granularity. Future work should therefore prioritise CLIP visual integration within the cross-model scoring framework, and apply quality-gating only to the noisier acoustic and OCR modalities.

6.3 Streaming and online inference

The current pipeline processes a complete video offline. A streaming variant would segment lecture content as it is being recorded, enabling real-time chaptering for live classes. This requires adapting the sliding-window fusion to operate on a causal context window and replacing the global threshold computation with an exponential moving average of past gap scores. The target latency budget for live chaptering is < 30 seconds of delay.

6.4 Cross-lingual extension

LECSEG currently targets English lectures. Extending to other languages requires: (1) replacing `all-mpnet-base-v2` with a multilingual text encoder such as `LaBSE` or `multilingual-e5-large`; and (2) using Whisper’s multilingual mode for transcription. Suitable evaluation data could be sourced from non-English MOOCs (e.g., Coursera’s French and Spanish tracks) or from university recordings in South and South-East Asian languages.

6.5 End-to-end fine-tuning

The current architecture uses frozen pre-trained encoders. Fine-tuning the text encoder jointly with the boundary predictor on a larger annotated corpus (e.g., constructed by scaling the LECSEG-30 annotation procedure to 200+ videos) could yield further gains, at the cost of reduced generalisability. A natural training objective is a soft version of the tolerance- F_1 metric, differentiable via a sigmoid boundary-probability output.

6.6 Larger and domain-tuned local LLMs

Replacing Llama-3.1-8B with a 70B parameter model or a domain-specific fine-tune (e.g., fine-tuned on lecture transcripts) could improve title quality and boundary-shift accuracy, particularly for highly technical content. Quantised models (GGUF 4-bit) make 70B inference feasible on a single 24 GB GPU with acceptable latency (≈ 5 seconds per boundary refinement call).

6.7 Multi-speaker lectures and panels

Panel discussions and multi-instructor courses introduce speaker turns as an additional boundary signal. Speaker diarisation (e.g., using `pyannote.audio`) could provide a speaker-

change feature that complements text embeddings, particularly for sections where the topic does not change but the speaker does.

6.8 User study on learning outcomes

All metrics in this thesis are information-retrieval proxies for segmentation quality. A controlled user study would directly measure whether learners locate target concepts faster, and whether their retention improves, when using LECSEG-generated chapters compared to flat video scrubbing or manually authored chapters. Such a study could also reveal which granularity (chapter vs. subtopic) is most useful for different learning tasks (review vs. first-watch).

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Appendix A

Dataset Details

A.1 Video list

Table A.1 lists the 30 videos in LECSEG-30. Full metadata, URLs, and timestamps are stored in `data/manifest.jsonl`, `data/gt/*.json`, and `data/gt_hier/*.json`.

Table A.1: LECSEG-30 video list, sorted by domain and video ID.

Video ID	Domain	Duration (min)	Chapters
9N1MxkbFhsc	Biology	42.8	23
AMl6E4cLrwE	Biology	45.6	13
KlVHq38KJU	Biology	38.8	15
NNnIGh9g6fA	Biology	57.2	16
oOya3cFmAMc	Biology	51.7	15
uO5k9xcD1gU	Biology	47.3	12
D8RRq3TbtHU	CS	114.5	12
HtSuA80QTyo	CS	53.4	9
JP7ITIXGpHk	CS	105.6	26
Rl0ludWTLxs	CS	149.5	21
TjZBTdzGeGg	CS	47.3	11
jGwO_UgTS7I	CS	75.3	7
oqRU2So6Z2Y	CS	135.6	12
7K1sB05pE0A	Math	51.5	13
fsLh-NYhOoU	Math	60.4	13
j0wJBEZdwLs	Math	34.7	8
lUUte2o2Sn8	Math	65.2	21
8yT4RZy1t3s	Philosophy	55.2	18
GR63MMai-fs	Philosophy	45.8	5
KNwMiydCYA4	Philosophy	63.4	25
MGyygiXMzRk	Philosophy	55.0	11
Qw4l1w0rkjs	Philosophy	55.1	34
S7TUe5w6RHo	Philosophy	63.5	29
Hy7ou5R_vjE	Physics	66.8	6
KOKnWaLiL8w	Physics	73.4	6
NK-BxowMIfg	Physics	66.0	5
Ns6GB4Dph9U	Physics	68.4	6
YdOXS_9_P4U	Physics	28.6	7
uK2eFv7ne_Q	Physics	73.8	6
zNVQfWC__evg	Physics	60.3	14
Total	5 domains	1951.4	419

A.2 Domain distribution

Table A.2: LECSEG-30 domain distribution.

Domain	Videos	Duration (hours)	Chapters
Biology	6	4.72	94
CS	7	11.35	98
Math	4	3.53	55
Philosophy	6	5.63	122
Physics	7	7.29	50
Total	30	32.52	419

A.3 Subtopic annotation guidelines

The following guidelines were given to annotators for the subtopic annotation pass after chapter boundaries were fixed from YouTube metadata:

1. **Boundary definition.** A subtopic boundary occurs when the speaker transitions from one coherent pedagogical sub-idea to another within a chapter.
2. **Minimum duration.** Each subtopic segment should be at least 60 seconds unless the source chapter is shorter than that.
3. **Number per chapter.** Annotators were encouraged to use 2–5 subtopics per chapter when the chapter length supported it.
4. **Title format.** Titles should be short noun phrases, preferably using the speaker’s own terminology.
5. **Uncertainty.** When unsure, annotators marked the best plausible transition. Disagreement was preserved for IAA analysis rather than hidden.

A.4 Inter-annotator agreement details

The final IAA report contains 30 videos and is stored in `data/gt_hier/iaa_report.json`. Mean chapter-level kappa is 0.5351 and mean subtopic-level kappa is 0.4257 at tolerance 1 sentence. Boundary F1 is 1.0000 for chapters and 0.7793 for subtopics.

Appendix B

Hyperparameters and System Configuration

All hyperparameters are stored in `configs/defaults.yaml` and can be overridden per experiment using Hydra. Every evaluation run records its exact configuration to `results/<run>/config.yaml`, ensuring full reproducibility.

B.1 Default configuration

B.2 Hardware and wall-clock runtimes

Experiments were run across the local Windows workstation and rented vast.ai GPU instances. Exact CPU/RAM values varied by run; GPU-backed runs used NVIDIA RTX-class hardware.

- CPU: local Windows workstation CPU for orchestration and light tests
- GPU: NVIDIA RTX 5090 for transcription; NVIDIA RTX 3090 for later embedding experiments
- RAM: instance-dependent; no run required model-parallel execution
- OS: Windows 11 locally; Ubuntu Linux on GPU instances

Approximate wall-clock times per pipeline stage for a 60-minute lecture:

The dominant cost is speech recognition; all other stages combined run in under two minutes per hour of video.

Table B.1: Key hyperparameters used in all reported experiments.

Component	Parameter	Value
ASR	Model	Whisper large-v3
	Beam size	1 (speed)
	Compute type	float16
	VAD filter	Enabled (500 ms min silence)
Text encoder	Backbone	all-mpnet-base-v2
	Embedding dimension	768
	L2-normalised	Yes
Visual encoder	Backbone	CLIP ViT-B/32
	Embedding dimension	512
	Keyframe rate	1 fps
Two-stage predictor	Stage-1 threshold c_1	0.5σ below mean
	Stage-2 threshold c_2	1.2σ below mean
	Context window w	3 sentences
Reliability weighting	Fusion mode	Entropy-based
	Modalities	Text, visual, prosody
LLM refinement	Model	Llama-3.1-8B (Q4_K_M)
	Snap window	± 2 sentences
	Title max words	8
Evaluation	Bootstrap samples	1,000
	Confidence level	95 %
	Significance test	Wilcoxon + Holm correction
	Tolerance	± 2 sentences
Random seed	All stochastic ops	42

Table B.2: Wall-clock time per pipeline stage for a representative 60-minute lecture.

Stage	Time (seconds)
Whisper large-v3 transcription	≈ 200
Sentence splitting (spaCy)	< 1
Text embedding (MPNet, CPU)	≈ 30
CLIP visual embedding (GPU, 1 fps)	≈ 60
RWF fusion + boundary prediction	< 1
LLM refinement + titling (8B Q4)	≈ 30
Total	≈ 320

Appendix C

Additional Results

C.1 Per-domain breakdown

Table C.1 reports the alignment-audited joint P_k /WindowDiff cross-model method, broken down by domain.

Table C.1: Per-domain evaluation of the current best official method.

Domain	Videos	$P_k \downarrow$	WD $\downarrow$	F1 $\uparrow$
Biology	6	0.3965	0.4009	0.0475
CS	7	0.3297	0.3338	0.0000
Mathematics	4	0.3793	0.3867	0.0000
Philosophy	6	0.3959	0.4073	0.0663
Physics	7	0.3665	0.3665	0.0000
All domains	30	0.3713	0.3764	0.0237

C.2 Additional benchmark rows

C.3 Representative cases

The per-video results expose three useful cases for interpreting aggregate metrics:

- Best case by P_k : oqRU2So6Z2Y (CS), $P_k = 0.1457$.
- Typical mid-range cases are documented in the stored per-video evaluation rows.
- Hard case: S7TUe5w6RHo (Philosophy), $P_k = 0.4975$.

Table C.2: Selected additional benchmark and diagnostic rows.

Method	$P_k \downarrow$	WD $\downarrow$	Source
TextTiling	0.6053	0.8978	eval_bge.json
C99	0.4219	0.4494	eval_bge.json
CosineSeg	0.4902	0.5392	eval_bge.json
KMeansSeg	0.6172	0.9986	eval_bge.json
BertSeg	0.4891	0.5403	eval_bge.json
BGE + divisive	0.3884	0.3956	eval_bge.json
Cross-model conservative	0.3713	0.3764	eval_alignment_sweep.json
LOO ExtraTrees selector	0.3588	0.3739	method_selector_experiment_trainrank_balanced.json
Per-video method oracle	0.2980	0.3280	method_portfolio_analysis.json
BERT-wiki zero-shot	0.4932	0.5397	eval_bert_wiki.json
Boundary classifier LOOCV	0.5803	0.9933	eval_boundary_clf.json

Together with the selector-choice audit in Table 4.7, these cases show that gains are concentrated in particular lectures rather than uniformly distributed across the benchmark.